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Evolution of Vertebrate Immunity

Review

Thomas Boehm

All multicellular organisms protect themselves against pathogens using sophisticated immune defenses. Functionally interconnected humoral and cellular facilities maintain immune homeostasis in the absence of overt infection and regulate the initiation and termination of immune responses directed against pathogens. Immune responses of invertebrates, such as flies, are innate and usually stereotyped; those of vertebrates, encompassing species as diverse as jawless fish and humans, are additionally adaptive, enabling more rapid and efficient immune reactivity upon repeated encounters with a pathogen. Many of the attributes historically defining innate and adaptive immunity are in fact common to both, blurring their functional distinction and emphasizing shared ancestry and co-evolution. These findings provide indications of the evolutionary forces underlying the origin of somatic diversification of antigen receptors and contribute to our understanding of the complex phenotypes of human immune disorders. Moreover, informed by phylogenetic considerations and inspired by improved knowledge of functional networks, new avenues emerge for innovative therapeutic strategies.

Introduction

All multicellular organisms rely on dedicated immune defense systems for their survival. Over a period of several hundred million years, a bewildering variety of defense strategies has emerged. Individual species typically employ many of these mechanisms simultaneously to achieve functional redundancy for one of an organism's most important facilities. The presence of multi-layered humoral and cellular immune systems is as vital for invertebrates as it is for humans. Indeed, our own highly sophisticated immune system exemplifies this functional diversity. Non-specific chemical defenses, such as those initiated by the broadly reactive C3 complement component, co-exist with cellular effector mechanisms, the latter best illustrated by cytotoxic T lymphocytes that exhibit exquisite target specificity for their peptide antigens to the level of single amino acids.

Historically, immune functions were conceptually divided into two separate, for the most part functionally independent, domains. So-called innate defenses were thought to rely on hard-wired programs of enzymatic and cellular activation, underlying stereotyped immune responses observed in invertebrates. On the other hand, the so-called adaptive responses of vertebrates, typified by the capacity for memory formation, were attributed to the function of different types of lymphocytes. This dichotomy of design was seemingly supported by successive appearance in evolutionary history, innate responses being present in invertebrates, and adaptive responses being the hallmark of vertebrate immune systems. However, recent research has led to the realization that innate immune systems also incorporate certain aspects of immune memory and that innate and adaptive immune functions are intimately linked. Many of these fundamental insights have emerged from comparative analyses, albeit from a limited number of model species. Hence, an evolutionarily informed view of immune functions promises not only to change our perception of the network structure of the immune system but also to lead to non-intuitive strategies aimed at interfering with states of immunodeficiency or immune deregulation.

In this review, I discuss recent developments to illustrate why research on animals separated from humans by several hundred million years of independent evolution provides crucial information for clinical research and modern medical practice.

Evolution of Innate and Adaptive Features of Immune Systems

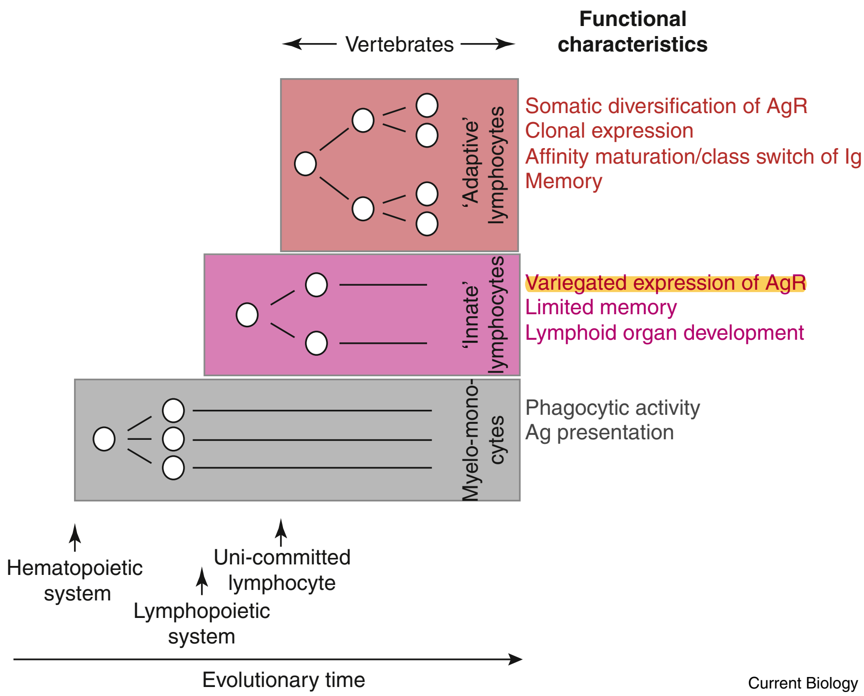
The immune systems of invertebrates [1–3], comprising the great majority of animal species on earth [4], are surprisingly complex and compare favorably with those of vertebrates [5–12] in terms of the multitude of mechanisms employed for immune defense, the plethora of immune-related molecules and the diversity of effector cell types. Insects, for instance, are capable of mounting cellular and humoral responses to invading pathogens; however, in contrast to vertebrates, no antigen-triggered proliferative responses have yet been demonstrated. Immune recognition in insects occurs directly or indirectly through germline-encoded pattern recognition receptors, such as Toll [3,13–15]. Immune responses employ multiple effector mechanisms: phagocytosis or encapsulation of pathogens; activation of specific enzymes, for instance the phenoloxidase cascade, to generate cytotoxic intermediates, such quinolones [3]; and expression of antimicrobial peptides [16].

Interestingly, some of the molecules implicated in host defense even exhibit somatic diversification, a phenomenon that was previously thought to be a distinguishing feature of vertebrate immune systems. In arthropods, for instance, this occurs by alternative splicing of the Dscam molecule, resulting in the formation of thousands of isoforms [17]; in mollusks, limited somatic diversification, possibly by a mechanism akin to somatic hypermutation, targets fibrinogen-related proteins (FREPs) [18], which are lectin-like molecules found in the hemolymph and capable of forming immune-complex-like aggregates [19]; in echinoderms, proteins of the Sp185/333 gene family appear to be somatically diversified [20]. Even more intriguingly, an infection can lead to increased, sometimes even cross-protecting resistance against pathogens other than those eliciting the primary response [21–24], and long-lasting effects on the composition of immune effector cell types, such that, for instance, mosquitoes are protected from *Plasmodium* parasite reinfection owing to increased numbers of granulocytes, a special subpopulation of hemocytes [25]. This kind of memory in invertebrates represents a kind of heightened state of alert and is referred to as 'trained immunity' [26] to distinguish it from the adaptive immunological memory in vertebrates that is based on clonal selection of immune effector cells [27]. Hence, as immune functions of

Department of Developmental Immunology, Max Planck Institute of Immunobiology and Epigenetics, Stuebeweg 51, D-79108 Freiburg, Germany.
E-mail: boehm@immunbio.mpg.de

Figure 1. Successive evolutionary emergence of myeloid and lymphoid cell types underlying the cellular arms of metazoan immune systems.

Initially, immune systems relied on the function of multi-purpose myelomonocytic cells arising from the hematopoietic cell lineage; phagocytosis was one of their major primordial autonomous immune defense functions; in vertebrate immune systems, they additionally function as antigen-presenting cells. Lymphocytes represent an innovation of vertebrates and occur in two forms. 'Innate' lymphocytes, such as natural killer cells, are characterized by variegated expression of germline-encoded antigen receptors, exhibit certain features of immune memory and participate in lymphoid organ development as so-called lymphoid tissue inducer cells. 'Adaptive' lymphocytes are distinguished by expression of somatically assembled antigen receptors, in a 'one receptor — one cell' fashion; such uni-committed lymphocytes probably emerged at a later point than 'innate' lymphocytes. Clonal expression of somatically assembled antigen receptors underlies further adaptive functions, such as affinity maturation and class switch recombination of immunoglobulins, and specific memory. Metazoan immune systems additionally possess humoral defense systems. AgR, antigen receptor; Ig, immunoglobulin; Ag, antigen.



invertebrates are studied in greater detail, it appears that they exhibit numerous hallmarks previously associated with vertebrate 'adaptive' immunity (Figure 1).

Hallmarks of Vertebrate Immune Systems

Vertebrates, a minor fraction of all known animal species, comprise two clades, the jawless vertebrates, consisting of approximately 100 species of hagfish and lamprey [28], and jawed vertebrates, encompassing tens of thousands of species, including mammals [4]. Previously, it was thought that the immune systems of vertebrates were characterized by the specific acquisition of adaptive immune functionalities, thus distinguishing them from the immune systems of invertebrates. However, as a consequence of the recent findings on invertebrates referred to above, and the findings on the innate arm of vertebrate immune systems described below, the functional distinctions between innate and adaptive immunity are becoming more and more blurred; indeed, the use of these terms — in both ontogenetic and phylogenetic contexts — is considered to create an unnecessary obstacle to conceptual progress [8,29,30]. This functional convergence is mirrored in the unexpected homologues in the molecular pathways involved in the development and function of blood cells of vertebrates and at least some invertebrate phyla [31]. Notwithstanding these developments, it is clear that vertebrate immune systems exhibit unique innovations, the most obvious being the presence of cells of the lymphoid lineage; furthermore, lymphocytes expressing somatically assembled receptors usually do so in a way that each lymphocyte is equipped with a single receptor [6], a unique advantage when it comes to purging the repertoire of self-reactive cells and generating immune memory, as discussed below (Figure 1).

Somatic Diversification of Antigen Receptors

Vertebrates use stunningly efficient somatic diversification processes to generate an anticipatory repertoire of structurally diverse antigen receptors that are expressed by two functionally distinct lineages of lymphocytes: antibody-producing B cells, and T cells that are associated with cytolytic and helper functions [6]. Interestingly, the types of genes that are subjected to somatic diversification in the two vertebrate clades are entirely different. Jawless vertebrates rely on leucine-rich-repeat-containing antigen receptors termed variable lymphocyte receptors (VLRs) [7,12]. VLRs are assembled from incomplete genomic elements by a gene conversion-like process [32] — a form of homologous recombination that is initiated by DNA double-stranded breaks and results in non-reciprocal transfer of genetic information. In the case of VLRs, this is presumably achieved by cytidine deaminases [33] in a lineage-specific fashion, with *CDA1* being involved in *VLRA* assembly, and *CDA2* in *VLRB* assembly [34]. The diversity of VLRs is generated by combinatorial use of variable numbers of leucine-rich repeat (LRR) elements and a patchwork-like assembly process [7,12]. B-like cells of lamprey express the VLRB isotype [34–36], which can be secreted in the same way as antibodies [37,38]. By contrast, the VLRA and VLRC antigen receptors [34,39,40] are expressed by two distinct T-cell-like lineages and are situated on the cell membrane, akin to a T-cell receptor (TCR) [34].

The repertoire of antigen receptors of jawed vertebrates is the result of both combinatorial and junctional diversity, collectively referred to as V(D)J recombination [41]. Junctional diversity leads to the generation of non-templated sequence elements, an important mechanistic difference between jawless and jawed vertebrates; the random sequences occur in the third complementarity-determining

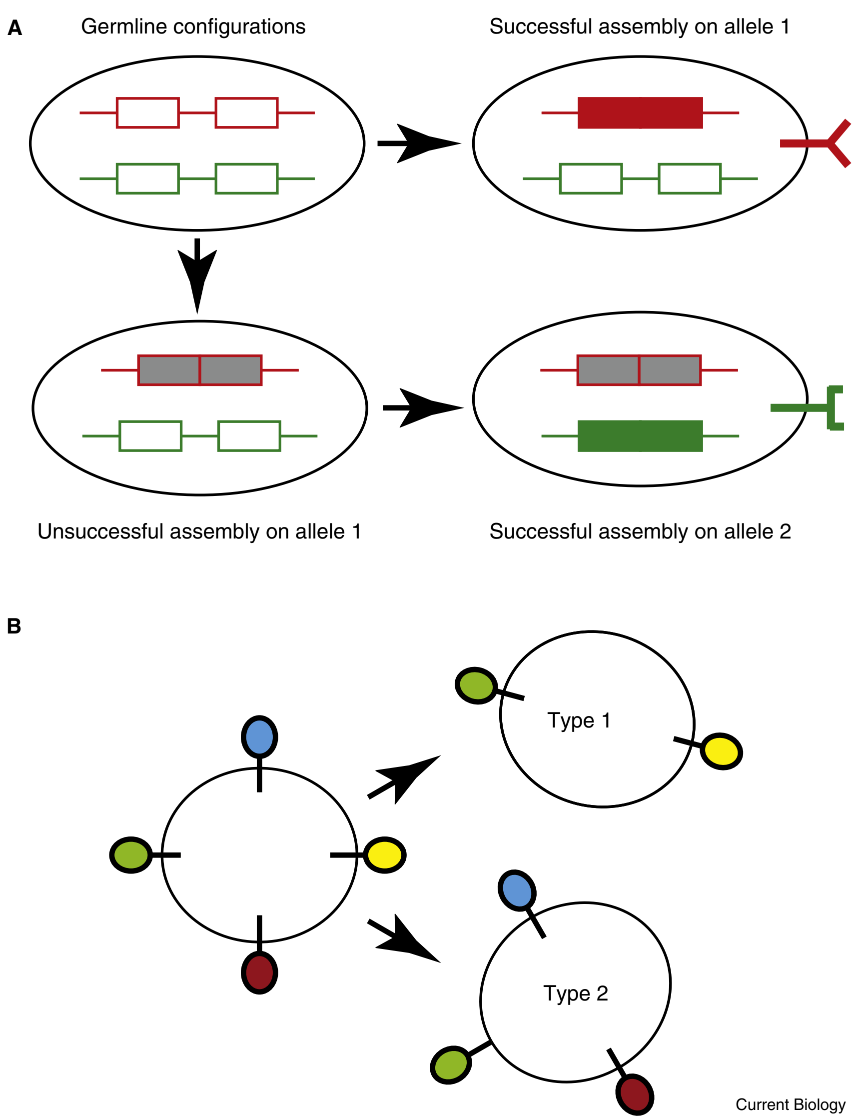


Figure 2. Distinct modes of antigen receptor expression in the immune system.

(A) The assembly of antigen receptor genes in somatic cells occurs in a step-wise fashion. If assembly on the first allele is functional, the assembly process is stopped, and the receptor encoded by this gene is expressed. If assembly on the first allele fails, an attempt is made on the second allele. If both attempts fail, a lymphocyte is presumed to abort further differentiation. (B) The expression of germline-encoded antigen receptors often occurs in a variegated fashion, in such a way that different cells express different subsets of the antigen receptor repertoire.

regions (CDR3) of immunoglobulins (Igs) and TCRs (encoding parts of the antigen-binding surface) and are the result of an error-prone non-homologous end-joining process, initiated by recombination-activating gene (Rag) family proteins [41]. Despite these mechanistic differences, the available evidence suggests that the degrees of diversity in antigen receptor repertoires achievable by somatic diversification in jawless and jawed vertebrates are essentially similar [36,42]. It should be noted that cytidine deaminases are also employed for antigen receptor gene diversification in jawed vertebrates, both during the generation of the primary repertoire of immunoglobulins and for their subsequent diversification during an immune response; indeed, they may form part of an ancestral mechanism of immune defense that targets foreign genetic material, establishing genetic and functional continuity between the immune systems of invertebrates and of the two branches of vertebrates [6,7,9–11].

Monoallelic Expression of Antigen Receptors

A second characteristic of somatically diversified antigen receptors in vertebrates is their expression mode [34,35,40,43,44]; because only one receptor is expressed per cell, each lymphocyte and all of its progeny (referred to as a clone) are endowed with a unique molecular and functional signature (Figure 2A). By contrast, the expression of germline-encoded additional receptors on lymphocytes, such as Toll-like receptors (TLRs), is generally not clonal, as lymphocytes expressing different types of somatically assembled antigen receptors might still express the same type(s) of germline-encoded receptors (Figure 2B). While it could be argued that somatic diversification already occurs in the immune systems of invertebrates, albeit on a much smaller scale than in vertebrates, clonal expression of antigen receptors possessing unique specificities has not yet been detected in invertebrates; it thus represents a distinguishing feature of vertebrate immune systems.

Note, however, that the 'one cell — one receptor' mode of expression can be considered as the extreme case of variegated expression of immune-related receptors that is known to occur frequently in the immune systems of both invertebrates and vertebrates: for instance, hemocytes — the principal immunocyte of arthropods [45] — express only a subset of the many thousands of isoforms of Dscam that are generated by alternative splicing [17]; likewise, each natural killer (NK) cell, a form of 'innate' lymphocyte of vertebrates responsible for the detection of infected or mutated cells of the body, expresses only a subset of receptors specifically surveying the presence of self-structures expressed by healthy cells [46]. Nevertheless, because expression of single specificities arising from somatic assembly is a feature of the lymphocytes of both jawless and jawed vertebrates, it must have conferred unique advantages to their immune systems, although it remains unclear why this is the case. However, it has been suggested that, because somatic diversification involves the risk of creating self-reactive receptors (that are generally eliminated or functionally incapacitated), expression of single specificities maximizes the efficiency with which the repertoire of self-tolerant cells can be generated in primary lymphoid organs [6]. Hence, it is possible that the clonal expression of single-specificity-type receptors that is observed for lymphocytes in the two groups of vertebrates arose by convergent evolution.

Memory Response

Clonal expression may also prove useful during the implementation of memory responses. Immune effector cells depend on trophic support for their maintenance; T cells, for instance, require the cytokine interleukin-7 (IL-7) and the presence of weakly reactive self-antigens [47]. However, because the limited supply of trophic factors essentially determines the absolute number of effector cells of a particular type that can be maintained by an organism, this resource must be carefully managed. When antigen receptor specificities are clonally distributed among different effector cells, an infection usually addresses only a small proportion of antigen-specific cells; hence, it is possible to increase their frequency considerably without significantly compromising the magnitude of the remaining naïve repertoire (required for possible subsequent infection by other kinds of pathogens). Considering that naïve and memory cells may have different requirements for maintenance [48,49], the advantage of this design becomes even more apparent. Because the magnitude of an immune response depends to a large extent on the frequency of relevant antigen-specific effector cells, a secondary immunological response is much more efficient than a primary response; moreover, when compared to that of naïve cells, the 'activated' state of memory cells enables faster deployment of anti-pathogen responses.

Immunological memory, long considered a critical and defining aspect of 'adaptive' responses in vertebrates, has now also been found to occur in an 'innate' context. For instance, NK cells employ a series of activating and inhibitory receptors to integrate a multitude of signals from the environment. During development, NK cells are equipped with different combinations of these receptors through the process of variegated expression [46]; once their net inhibitory potential — the result of a process referred to as licensing — is overturned by viral infection or an encounter with tumor cells, the activated fraction in the NK repertoire can transit to a memory-like state [50]. Hence, a subsequent memory response exhibits features of antigen specificity. For instance, during infections with mouse cytomegalovirus, NK cells expressing the activating receptor Ly49H recognize the viral m157 glycoprotein, undergo expansion, contraction and memory cell formation and, upon re-infection, demonstrate enhanced cytolytic function and cytokine production, the classical signature of an adaptive memory response [50].

An Evolutionary Enigma

The two groups of extant vertebrates share many principles of immune system design and development (Figure 1), indicating that these key features must have been present in their common ancestor [6]. However, it seems likely that wholesale somatic diversification of antigen receptors was not yet present in the hypothetical ancestral vertebrate. Rather, at some point after the divergence of the two morphologically (and presumably also ecologically) different groups of vertebrates, their respective ancestors must have experienced a similar type of selection pressure that eventually led to the evolution of mechanistically distinct but functionally similar somatic diversification of the antigen receptors expressed on lymphocytes. What could this evolutionary force have been?

Origin of Somatic Diversification of Clonally Expressed Antigen Receptors

The morphological complexity of vertebrates is generally believed to be greater than that of invertebrates and it has been suggested that the increase of morphological and functional complexity was unusually fast during the early stages of vertebrate evolution [51–53]. This rapid progress might have resulted in a greater variety of habitable ecological niches per species and consequently in a heightened susceptibility to infection by a wider spectrum of pathogens. The exposure to the diverse requirements of their expanded habitats may thus have provided the evolutionary force for more sophisticated immune defenses in early vertebrates [54]. Thus, while increased morphological complexity might have increased vulnerability to pathogen attack, it also afforded unprecedented opportunities for elaborating novel types of immune effector cells that could eventually be integrated into multi-dimensional regulatory circuits of immune defense (Figure 1).

Host Adaptations to a Super-Organismic Lifestyle

An alternative, but not mutually exclusive hypothesis posits that the vertebrate-specific innovations in the immune system provided the basis for the evolution of complex mutualistic relationships between the vertebrate host and beneficial microbes [55] (Figure 3). Mutualism between host and microbiome greatly expands the metabolic facility of the resulting super-organism (a sociological term recently popularized by work on insect societies [56]), to the benefit of both parties [57]. Indeed, numerous examples exist for invertebrates to support this notion, which also applies to humans, as evidenced, for instance, by the production of vitamins by colonic bacteria as a component of complex networks of inter-specific and intra-specific metabolic interactions [58]. Hence, if metabolic proficiency and/or lifestyle of a vertebrate depend to a greater extent on the synthetic facilities of the microbiome than do those of their non-vertebrate cousins, conditions conducive to species-rich microbiomes might have been favored in the early stages of vertebrate evolution.

In considering the interactions between host and microbiome, it is important to distinguish between resident (autochthonous) flora and passenger microbes (allothonous flora, i.e. food or pathogens). Whereas the former would be the target of evolving mutualistic interactions, undesired inhabitants potentially necessitated the ability to initiate immune defenses. Could it be possible then that an immune system employing structurally diversified antigen receptors facilitated increased species-richness in autochthonous microbial communities, for example, in the intestine? It is important to note, however, that the elaboration of structurally diverse antigen receptor repertoires by consecutive rounds of gene duplication and diversification would have been severely constrained by the resulting rapid increase in genome complexity, with its attendant cost. By contrast, somatic diversification of antigen receptor genes would have offered an efficient (and elegant) solution to achieve vastly diverse repertoires while requiring only a minimally increased coding capacity of the genome. Is there evidence to support this idea? Indeed, microbiomes of invertebrate 'super-organisms' are much more arid in terms of species

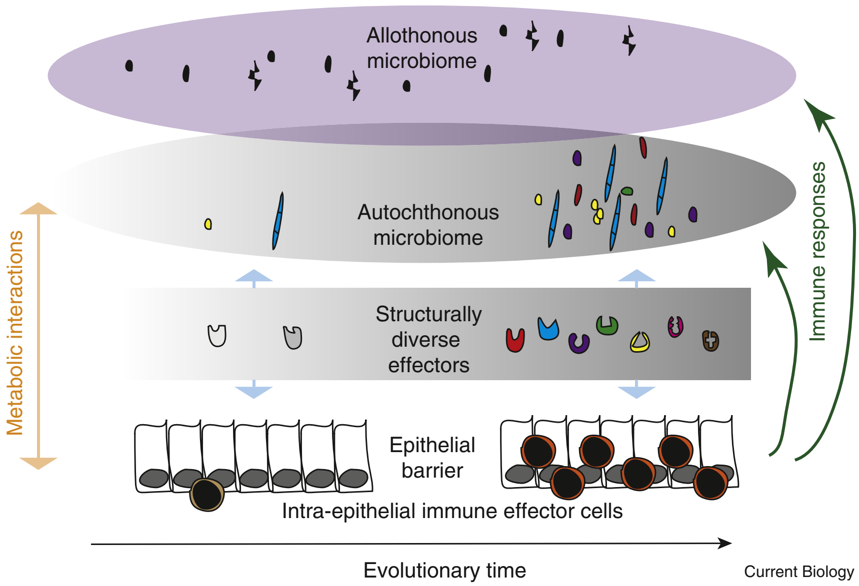


Figure 3. Schematic illustrating the role of a diversified antigen receptor repertoire in mediating the elaboration of a species-rich microbiome living on and in vertebrates.

It is proposed that the structural diversity of antigen receptors generated by intra-epithelial immune effector cells scales with the species diversity of the autochthonous (resident) microbiome; this in turn establishes stable metabolic interactions with the host organism. At the same time, antigen receptors additionally interact with transient (allothonous) components of the microbiome; these immune responses eventually regulate its composition.

diversity and metabolic capacity [59] than those living on and in vertebrates [60]. To examine how general these findings are, it will be particularly important to examine the diversity of autochthonous intestinal taxa in many different animal species living in aquatic environments, for example, evolutionarily ancient deuterostome species, such as acorn worms and sea urchin, in addition to evolutionarily younger chordates, such as cephalochordates and tunicates, and to compare them with those in lower vertebrates such as lamprey, hagfish and cartilaginous fish.

Sustaining a Species-Diverse Microbiome

The selective advantage of increasing antigen receptor diversity with respect to the species-richness of microbiomes (Figure 3) is illustrated by the apparent role of secreted antibodies, such as IgA in mammals, in the maintenance of microbial homeostasis on mucosal surfaces; defective structural diversification of secreted antibodies is associated with dysbiosis, which is characterized by generally lower species diversity and an ‘unhealthy’ composition of the microbiome [61–70]. If the hypothesis that an immune system employing somatically diversified antigen receptors was required for the establishment of diverse microbiomes is indeed true, then several predictions can be made. For instance, it would be expected that evolutionarily older vertebrates, such as fish, amphibians and reptiles, also employ secretory immunoglobulins [71,72] for this purpose: indeed, in teleosts, the IgT (also known as IgZ) isotype is secreted into the intestinal lumen, where it coats the resident microbes [71]. This functional flexibility is mirrored in the IgA-deficient state of mammals, in which IgM can functionally replace secreted IgA [73]. Remarkably, a pentameric form of VLRB antibodies has been detected in lamprey [37]; however, it is unknown whether the structural analogies to the secreted pentameric form of IgM also extend to functional aspects, that is, the secretion into the intestinal lumen. Another expectation is that the correlation between the adaptive facilities of the immune system and the diversity of associated microbiomes also holds for lower vertebrates. This prediction can be tested using a series of recently described immunodeficient teleost fish [74]; of particular interest are fish that lack either all lymphocytes [75] or only the B-lymphocyte lineage associated with expression of IgT/Z [76]. It is also possible that chordates already invented the principle of immune-mediated microbial management by secreting structurally diverse immune-related molecules.

It has been suggested that variable-region-containing chitin-binding proteins (VCBPs) expressed in the intestinal tract of living chordates, such as amphioxus and tunicates, are involved in shaping resident microbial communities [77,78]. In amphioxus, several VCBPs are encoded by diverse, non-recombining and haplotypically variable alleles [77], whereas the degree of polymorphism in other VCBPs of amphioxus and in those of the ascidian *Ciona intestinalis* is much less pronounced [78]. If no other polymorphic immune-related receptors were secreted into the gut lumen in these species, the prediction would be that the autochthonous commensal communities in amphioxus are more diverse than in *C. intestinalis* and that the higher degree of diversity in some VCBPs of amphioxus reflects adaptive changes to specific (presumably amphioxus-specific) antigenic challenges from the allothonous flora. In addition, the diversity of intestinal microbiomes in different tunicate species (known to be quite variable with respect to their super-organismic metabolic capacities [79]) should scale with a species-specific degree of VCBP polymorphism.

Hence, in addition to immune defense in the classical sense, the elaboration and maintenance of a species-rich community of commensal microbes through diverse immune effector molecules would have other benefits: it would establish unfavorable conditions for invading micro-organisms, and in this way a form of non-specific protection against infection, and, as a result of the ‘buffering’ effect with respect to quantitative and qualitative perturbations of the commensal flora, it would at the same time ensure stability of metabolic capacities.

A Potential Drawback of the Super-Organismic Lifestyle

If the requirement for sophisticated host–microbe mutualism associated with vertebrate evolution was indeed the main force underlying the emergence of somatic diversification of antigen receptors, then there was a price to pay for advanced metabolic functionalities and the other benefits of the super-organismic life-style. As a result of the vast diversity of antigen receptors generated by somatic diversification, their inadvertent reactivity towards self-structures potentially initiates self-destructive immune responses [80]. While Darwinian selection weeds out potentially self-reactive forms of germline-encoded antigen receptors (such as TLRs), somatic diversification necessitates innovative solutions [5]. These are provided in the form of somatic quality control that is orchestrated by the microenvironment of the emerging primary lymphoid organs within which lymphocytes develop [81]. For instance, in the thymus, the site of T-cell development [6], complex cellular and molecular mechanisms underlying presentation of self-antigens [82] serve the purpose of dealing with overly self-reactive lymphocytes [83]. Hence, the emerging super-organisms traded investment into sophisticated mechanisms of somatic diversification, quality control of antigen receptors and maintenance of a functional immune system against the benefits of increased metabolic capacity and non-specific pathogen protection. In this view, the residual level of autoimmunity observed in vertebrates is perhaps not the result of faulty design but rather the optimal solution to an essential aspect of super-organismic lifestyle. If true, the mechanisms of somatic quality control should have evolved (or been co-opted from more ancient facilities) early in the evolution of vertebrates [5,6,84].

Disorders of Host Defense

Infants with congenital neutropenia [85], X-linked agammaglobulinemia [86], and severe combined immunodeficiency (SCID) [87] exemplified the genetic basis of immunodeficiency syndromes. Since then, the mutations underlying many of these primary immunodeficiency disorders have been identified and the Mendelian basis of clinically milder forms of impaired immune functions uncovered [88,89]. Disorders of immune regulation manifest themselves as autoimmunity, for example, following mutations in the autoimmune regulator (AIRE) gene [90]; in other cases, exceptional susceptibility to specific infectious diseases results in, for instance, chronic mucocutaneous candidiasis as a result of mutations in genes encoding the cytokine IL-17F and its receptor IL-17RA [91]. Indeed, in many cases, the discoveries of disease-causing genetic aberrations in human patients highlighted general phenomena, for instance, the role of apoptosis [92,93] and regulatory cellular interactions [94] in immune homeostasis. In humans, inborn errors of immunity are probably best defined using clinical criteria, laboratory parameters being, by necessity, in constant flux as technology develops. Impaired immune functions can lead to life-threatening infection and/or to autoimmunity; in the pre-medical era, the former was almost certainly a more significant problem than the latter.

Because of the enormous toll claimed by infections, it has even been argued that most humans are inherently immunodeficient [88], emphasizing the notion that, during the evolutionary arms race between host and pathogens, the latter constantly uncover new types of host susceptibility. As a result, ongoing host–pathogen interactions manifest themselves in signatures of Darwinian selection in immune-related host genes. For instance, purifying selection is a prominent feature of human type I interferons, indicating their essential roles and hence their lack of functional redundancy in the immune system [95]. Signs of adaptive evolution, for instance, of genes encoding the pattern recognition machinery [96], indicate how environmental factors, such as different pathogen spectra, are capable of shaping immune functions in certain populations. Such comparative studies shed light on the evolutionary conflict between host and pathogens, for instance, the genetic constraints on the compatibility of immune-related signaling molecules and their receptors that are also used by viruses to gain access to immune cells [97,98].

Animal Models Supporting Human Immunology Research

The 'experiments of nature' manifesting themselves in genetic disorders of immune system development and function have been replicated in forward genetic screens, predominantly in mice [99,100], and also, albeit to a lesser extent, in lower vertebrates, such as zebrafish [74,101] and medaka [74]. These phenotype-driven screens, together with comparative phylogenetic analyses, using the ever-increasing number of complete vertebrate genomes, and experimental perturbations in animals situated at various points along the evolutionary trajectory, are beginning to reveal the evolutionarily conserved core structures of genetic networks regulating immune system development and function [74]. To this end, the imminent elucidation of the genome sequences of jawless and cartilaginous fishes will provide critical additional information for the validation of hypotheses concerning the structure of the immune systems of lower vertebrates.

Comparative approaches have always been considered to be important for immunological research. Pioneering work on amphibians (for instance, [102]) and birds (for instance, [103–106]) established key principles of lymphocyte and lymphoid organ development and mechanisms of antigen receptor diversification. Indeed, these animal models continue to prove their worth. For instance, versatile transgenic procedures have been developed for the *Xenopus* model, which, together with a well-developed armamentarium of specific monoclonal antibody reagents and the powerful procedures for tissue manipulation, reaffirm the importance of amphibians for immunological research [107]. Likewise, the unique facilities afforded by the DT40 chicken cell line, which enables efficient gene inactivation owing to a high rate of homologous recombination, have led to significant insights into immune-related processes of DNA modification [108].

Whereas the mouse model has risen to unchallenged prominence, particularly due to the advantages of pre-determined genetic modifications that are possible in this species, additional animal models have slowly gained ground in immunological research. In particular, the use of genetically tractable fish species, such as the zebrafish, with unique opportunities for time-lapse recording of hematopoietic and immune cell development (for instance, [109]) and for the monitoring of ongoing immune responses [110], has led to major advances. Indeed, fish are a unique group of vertebrates distinguished by their extraordinarily diverse lifespans and ecology, thus promising to provide essential information on immune function. For instance, the recent discovery that a long-lived marine fish, the Atlantic cod, lacks the entire machinery of major histocompatibility complex (MHC) class II-related antigen presentation has provided insight into unprecedented alternatives of immune system design distinguished by increased numbers of MHC class I genes and genes encoding TLR-like pattern recognition

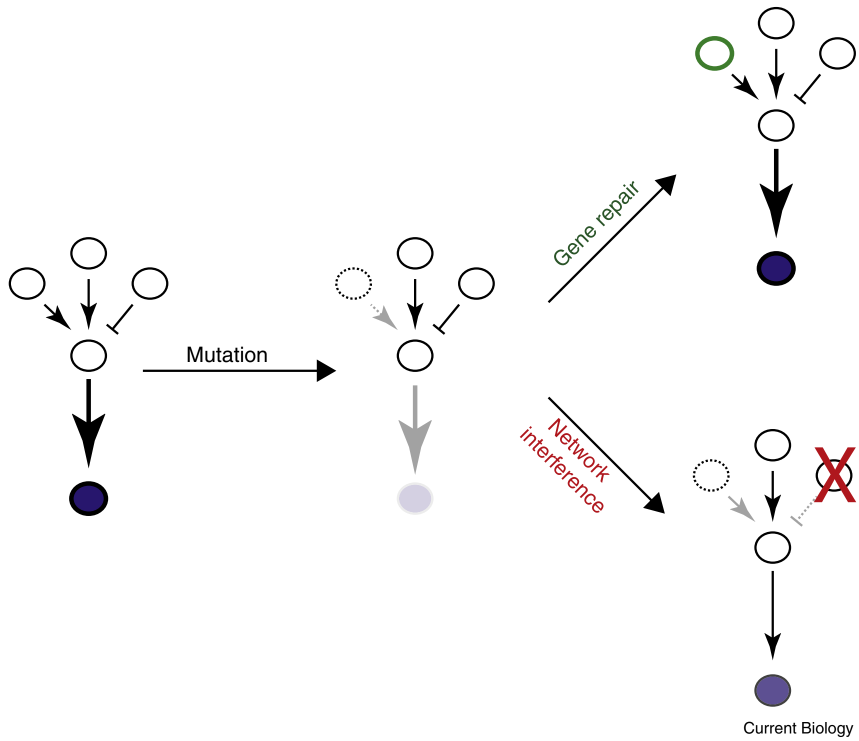


Figure 4. Network-inspired therapy.

In the genetic network depicted on the left, two positive effectors (arrows) and one inhibitory effector (T-shaped symbol) impact on the activity of a downstream target that itself positively regulates the output function (bottom circle). Following the inactivation of one positive upstream regulator by mutation, network output diminishes. Output activity is restored by correction of faulty gene function (top right panel). A precise understanding of the network structure suggests non-intuitive therapeutic strategies, here exemplified by removing the inhibitory function of an upstream regulator (bottom right panel).

receptors [111]. As another example, it was recently suggested that IgD, an immunoglobulin found in mucosal secretions as well as in the blood, plays an evolutionarily conserved role in systemic immunity. IgD binding to cell-surface receptors of basophils provides these innate cells with information about the antigenic composition of, for example, the respiratory tract and enables them to trigger rapid innate responses [112]. Supporting the reawakening of comparative vertebrate immunology is the growing importance of genetically tractable invertebrate models, contributing important information concerning evolutionarily ancient components of hard-wired immune defenses; instructive examples are the discovery of pattern recognition receptors in the fly [3] and histocompatibility systems in ascidians [113].

If the biological diversity of vertebrates is to be harnessed for the identification (and perhaps future therapeutic exploitation) of core components of the immune system, surrogate experimental systems to study particular features of genetically intractable species, particularly of fish, are urgently needed. In some instances, the development of interspecific transplantation models may offer an interesting path to follow. For instance, zebrafish carrying a loss-of-function mutation in the gene encoding the hematopoietic transcription factor c-Myb were found to lack definitive hematopoiesis [75]. These fish are currently being tested for use as universal recipients for cell and tissue transplants; this is of considerable interest as the hosts do not require conditioning prior to transplantation [74] and are themselves amenable to genetic manipulation.

Treating Immune Dysfunction

The understanding of immune regulatory networks has inspired numerous attempts to interfere with states of dysregulation or malfunction and the lives of countless patients suffering from immunological disorders have already been transformed by the clinical application of new strategies.

Clinical Application of Immunological Insights

A multitude of treatment strategies has been developed to alleviate immune dysfunction, some of which are illustrated below. Despite its conceptual simplicity, the use of polyclonal IgG preparations to treat patients with primary immunodeficiencies has resulted in dramatic clinical improvements [114], illustrating the contribution of a diverse immunoglobulin repertoire for immune homeostasis. The development and use of inhibitors of the potent pro-inflammatory cytokine IL-1 to treat an etiologically heterogeneic group of autoinflammatory syndromes, characterized by episodes of fever in the absence of activated T cells and autoantibodies and presumably driven by abnormal responses of innate effector cells [115], has met with extraordinary clinical success [116]. This treatment strategy is validated (and indeed mechanistically explained) by the fact that IL-1 and related cytokines are the end-product of a stereotyped inflammatory response mediated by the so-called inflammasome, a multiprotein complex sensing the presence of a variety of microbial, stress- and damage-related signals [117]. A similar clinical success has been achieved with antibodies neutralizing the activity of tumour necrosis factor (TNF), a cytokine that is expressed at very high levels in rheumatoid joint tissue [118].

While the above examples concern individual effector molecules of the immune system, it has also been possible to target its cellular components. For instance, the depletion of B cells with antibodies against the B-cell lineage-specific CD20 surface molecule benefits patients with autoimmune disorders, such as multiple sclerosis [119], and those suffering from certain B-cell malignancies [120]. As all immune effector cells are descendants of the hematopoietic stem cell, the clinical course of primary immunodeficiencies can also be modulated by hematopoietic cell transplantation [89], although allogeneic bone marrow transplantation is associated with significant treatment-associated mortality. Some disorders of immune regulation may also improve upon adoptive cell transfer; an example for this strategy is the use of *ex vivo* expanded antigen-specific T cells in patients suffering from reactivated cytomegalovirus disease

[121]. In recent years, clinicians have begun to employ gene therapy to treat monogenic immunological disorders, using viral vector systems to transduce hematopoietic stem cells to avoid problems with allogeneic transplantation procedures; however, although a functional reconstitution of genetic defects was achieved in the majority of patients, severe side effects associated with vector-associated insertional mutagenesis continue to limit the clinical usefulness of this approach [122]. As a conceptual alternative, the use of sequence-specific designer nucleases is advocated as a novel means for the targeted correction of genetic lesions [123,124] in the relevant tissue.

Future Strategies to Interfere with Failing Immunity

In addition to those strategies discussed above, recent research hints at the possibility of developing alternative strategies to alleviate failing immune functions. As the genetic networks underlying the development of major cell lineages and organs in the immune system are gradually being revealed, comparative analyses using different vertebrate species promise to help the delineation of core elements of such networks. It may then be possible to at least partially normalize the output of such networks, not by restoring the function of faulty components but rather by adjusting the resulting imbalance through interference with other network components (Figure 4). For instance, while *Casp8*-deficiency causes embryonic lethality in mice, presumably owing to aberrations of intracellular signaling pathways in immune cells, simultaneous disruption of *Rip3*, a gene functioning downstream of *Casp8*, prevents lethality and restores initial immune competence [125]. Thus, whereas the increasing information on genetic interactions of immune-related genes [126–128] point to the interdependence of gene functions, this information might be exploited to develop non-intuitive treatment strategies to alleviate the phenotype of certain genetic defects. In addition, information on network structure could be used, for instance, to manipulate lineage fate decisions or to create artificial microenvironments for hematopoietic cell differentiation. With regard to the latter and using dysfunctional thymic epithelial cells as a starting point, hematopoietic functions in mice have been engineered *in vivo* by combinatorial expression of a limited number of factors supporting attraction, proliferation and specification of progenitor cells [129]; in this instance, the selection of factors for the reconstitution was guided by information on the physiology of single mutants in various species but also by knowledge about their evolutionary trajectory. Hence, phylogenetically informed and network-inspired strategies hold great promise for the treatment of immune system disorders.

In addition, unprecedented strategies for immune interference could be developed by exploiting examples of evolutionary remodeling of effector and signaling pathways, such as those observed in the innate immune system of the basal chordate amphioxus [130]. Furthermore, given the intricate co-evolutionary relationship and functional interdependence of host and microbiome, it might also be possible to manipulate the microbiome in order to affect the host immune system. While this appears to be an obvious strategy to curb inflammatory bowel disease, it is also possible that such interference would benefit patients with systemic immune dysregulation, such as asthma and allergies [58,131].

Conclusions

The recent discoveries of alternative antigen receptor systems in jawless vertebrates and of the unusual structure of immune systems in certain fish species have led to a renewed interest in comparative immunology. Although most immunological studies still focus on mammals (including humans), it is anticipated that analyses of lower vertebrates will provide additional unique insights into the core structure of adaptive immune systems, the modes of interaction between their essential components, and the functional and structural adaptations to different lifestyles and occupancy of diverse ecological niches. This will lead to the identification of many more species-specific idiosyncrasies, each of which will provide a potential blueprint for how failing components might be compensated through augmentation and/or rewiring of the genetic networks underlying immune functions. In this way, comparative immunology may well become a major driver for the development of non-intuitive treatment strategies of human immunodeficiency and/or immune deregulation.

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